

BlzPilot AI

Operations & Administration Manual

Retro-Future MSME Operations Copilot

TECHNICAL SPECIFICATION & DOCUMENTATION

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Target Platform: MSMEs, Wholesale & Retail Operations

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1. System Architecture Overview

BlzPilot is an all-in-one operations cockpit designed for Micro, Small, and Medium Enterprises (MSMEs). It automates bookkeeping, inventory, workforce wage settlements, and provides real-time reporting via a modern, high-fidelity developer dashboard styling.

The system runs on a modern decoupled architecture:

1. React Frontend (Vite + TailwindCSS v4)

Renders charts (Recharts), manages visual visual themes, local user states, and handles Google Sign-In with Firebase client configurations.

2. FastAPI Backend (Python)

Coordinates API routes, invokes Google Gemini models for invoice OCR extraction & daily summaries, and compiles professional invoice PDFs.

3. Dual Database Persistence Layer

Supports MongoDB via connection string (DATABASE_URL) for production operations. If unconfigured, automatically falls back to an offline local file database (backend/data_db.json).

2. Required API Credentials & Secrets

To run 100% of the copilot features, the following environment secrets and files must be configured:

API Key / Config	Location	Purpose
GEMINI_API_KEY	.env / backend/.env	Powers Gemini 2.5 Flash OCR parsing & advice.
firebase-applet-config.json	Project Root	Authenticates Google Sign-In & backup sessions.
DATABASE_URL	backend/.env	MongoDB connection (Optional fallback to local file).

3. Core Features & Operations Guide

A. Professional Invoicing & Receipt OCR

Under 'Billing & Invoices', operators can generate invoices manually or upload an image snapshot/PDF receipt of a vendor bill. Gemini extracts structured data automatically. Upon submission, the system generates a printable PDF receipt and commits the transaction directly to the financial statement ledger.

B. Inventory safety Watchlist

Under 'Inventory', products can be registered with active cost, pricing, stock levels, and restock safety limits. If quantities drop below the safety limit, the dashboard automatically issues a reorder alert.

C. Installer Workforce & Payroll

Under 'Staff & Wages', operators can log worker details, daily attendance, and assign work hours. Payroll balances are calculated automatically and can be settled on-demand, adding the payout log as an outflow transaction.

4. The Purpose & Benefits of PDF Formats

Standardized PDF documents generated by BlzPilot serve as critical business artifacts:

- Immutable Record Keeping: Static invoice files act as historical audit trails for tax compliance.
- Portability & Distribution: Easily shared with customers via WhatsApp, email, or Slack.
- Offline Accessibility: PDF files are stored inside 'backend/static/invoices/' and remain accessible even without an active internet connection.